PROJECT SYN

Cyber-Resilient Disaster Healthcare Command Center

Powered by Dual Bare-Metal IBM LinuxONE III (s390x) Mainframes // IBM Z Datathon 2026

Track: Real-Time AI for Critical Decisions / AI Secured | Category: AI Innovation Excellence

Engineering Team: Yuvraj Lokhande (Lead Architect), Palak, Anshika, Nidhi, Ritesh, Bhavya

// URBAN PROBLEM STATEMENT // IBM Z DATATHON 2026

The Crisis: Extreme Monsoon Inundation & Systemic Fragility

Mumbai Severe Monsoon Downpour & Coastal Sea Surge Emergency

1. Electrical Substation Cascades

GRID FAILURE: SUBSTATIONS TRIP

- Severe monsoon downpours flood low-lying Dadar & Lalbaug basins.
- BEST 22kV substations trip, cutting municipal power to major hospitals.
- Trauma wards drop onto finite diesel backup with zero real-time fuel visibility.

2. Arterial Corridor Submersion

CORRIDOR BLOCK: CRITICAL

- Hindmata & Milan subways submerge under standing floodwaters.
- Ambulances carrying critical ICU patients trapped in dead-end gridlocks.
- Standard consumer GPS apps route vehicles into drowned underpasses.

3. Adversarial Cyber Poisoning

TELEMETRY VULNERABLE

- Unauthenticated IoT telemetry risks corrupting hospital bed registries.
- Spoofed packets threaten to divert emergency blood convoys to closed facilities.
- Undersea fiber cables flood, cutting off centralized commercial cloud AI.

// ARCHITECTURAL BLUEPRINT // IBM Z DATATHON 2026

The 3 Load-Bearing Pillars of Project Syn

Strict Domain Isolation: Edge Verification, Deterministic Routing & Tamper-Evident Ledger

Pillar 1: Trusted Telemetry

CPACF GATE: < 0.24 ms

- IBM z16 CPACF Hardware Gate.
- Wire-speed HMAC-SHA256 in <0.24ms.
- Monotonic anti-replay ring buffer.
- Zero-Trust: Drops spoofed packets before AI inference or database write.

Pillar 2: Safety-Bounded AI

SNAP ML SPEED: < 0.8 ms

- IBM Snap ML Random Forest (<0.8ms).
- Deterministic Dijkstra Pathfinder.
- Dynamic Ambedkar Flyover bypass.
- Tri-Sovereignty AI: Cloud Gemini 3 -> Local Big-Endian Granite 1.5B s390x.

Pillar 3: Dual-Node Mesh

QUEUE: 1,200+ req/sec

- Dual LinuxONE bare-metal instances.
- FastAPI asyncio.Queue: 1,200+ req/s.
- Chained SQLite WAL block ledger.
- Cross-node P2P hash pinning between KEM Apex and Tata Memorial Hub.

// ENTERPRISE FEATURES // IBM Z DATATHON 2026

11 Core Capabilities Engineered for Urban Survival

Complete Production Implementation Across Frontend, Logic, and Mainframe Core

1. CPACF-WireGate
- Native IBM z16 KMAC crypto.
- <0.24ms HMAC-SHA256 verification.

2. Snap-Inference Engine
- IBM Snap ML tree inference in <0.8ms.
- Ward exhaustion anomaly scoring.

3. Dijkstra-Pathfinder
- Deterministic green corridor routing.
- Elevated Ambedkar flyover bypass.

4. Tri-Sovereignty AI
- Gemini 3 Flash -> OpenRouter ->
- Air-gapped IBM Granite 1.5B on s390x.

5. Trilingual Triage
- Autonomous clinical disaster briefs.
- Native English, Hindi, and Marathi.

6. Tactical Floorplans
- Interactive GIS map of 35 hospitals.
- Real-time ICU and generator bus plans.

// AI ARCHITECTURE // IBM Z DATATHON 2026

Tri-Sovereignty AI Cascade: 100% Offline Autonomy

Seamless Intelligent Failover across Connected, Sovereign, and Degraded Modes

Tier 1: Cloud Primary

TIER 1: GEMINI 3

- Google Gemini 3 Flash / 2.5 Pro.
- 1,000,000 token context window.
- Handles complex multi-hospital logistics across Greater Mumbai.
- Active in Connected Mode.

Tier 2: Rate Failover

TIER 2: < 0.4s FAILOVER

- Gemini 3.1 Pro / Flash-Lite.
- Auto-engages in <0.4s on HTTP 429 quota exhaustion.
- High-speed tactical triage summaries.
- Zero interruption to dispatch.

Tier 3: Multi-Cloud

TIER 3: MULTI-CLOUD

- OpenRouter Free Cascade.
- Meta LLaMA 3.3 70B, Qwen 2.5 72B, DeepSeek R1.
- Zero-cost multi-cloud redundancy.
- Active in Sovereign Mode.

Tier 4: Air-Gapped

TIER 4: AIR-GAPPED

- IBM Granite 3.0 1.5B (s390x).
- 1.18 GB INT4 quantized SLM.
- Executes natively in LinuxONE Big-Endian memory (~12ms/tok).
- 100% offline survivability.

// ZERO-TRUST SECURITY // IBM Z DATATHON 2026

CPACF Hardware Gate & Anti-Replay Ring Buffer

Wire-Speed Cryptographic Protection Against Malicious Telemetry Injection

Pre-Inference Cryptographic Gate

CRYPTO SPEED: < 0.24 ms

- Native IBM z16 CPACF hardware KMAC acceleration.
- HMAC-SHA256 signature verification in <0.24 milliseconds.
- Monotonic nonce cache rejects replayed packets.
- Zero CPU penalty compared to software-only cryptographic routines.
- Verified across 10,000-packet stress test in JupyterLab Notebook 04.
- 100% of injected counterfeit packets blocked at the wire.

Chained Forensic Audit Ledger

INGESTION: 1,200+ req/s

- FastAPI asyncio.Queue absorbs 1,200+ packets/sec shock loads.
- SQLite Write-Ahead Logging with SHA-256 block chaining:
- Block_n = SHA256(Block_{n-1} + Nonce + TelemetryPayload).
- Produces a tamper-evident forensic ledger for incident investigations.
- Continuous cross-pinning between Node 1 (KEM) and Node 2 (Tata Memorial).
- Survives complete node loss with instant peer failover.

// OPERATIONS CENTER UX // IBM Z DATATHON 2026

Tactical Floorplans & Sentinel-ATS Display Adapter

Designed for Emergency Projector Walls, Command Monitors & Paramedics

Interactive Tactical Floorplans

35 HOSPITALS INTERACTIVE

- Clicking any of Mumbai's 35 hospitals on the Leaflet map opens its layout.
- Displays real-time ICU ward occupancy, oxygen pipeline pressure, and diesel gensets.
- Visual color codes: Green (Nominal), Yellow (Triage Strain), Red (Exhausted).
- Provides field commanders with instant micro-level hospital visibility.
- Automated peer rebalancing dispatches patients to nearest open facility.

Sentinel-ATS TV Kiosk Mode

TV KIOSK ADAPTER

- One-click TV Kiosk Fullscreen Mode with tactical night Carbon theme.
- Adaptive layout: h-[72vh] min-h-[540px] max-h-[820px] fits all screen sizes.
- Anti-scroll-trap map interaction: Click-to-activate mousewheel zoom.
- Floating tactical spatial controls: Dark Matter, Voyager Roads, Positron Light.
- Quick 'Center Mumbai' button instantly refocuses on metropolitan coordinates.

// SCIENTIFIC RIGOR // IBM Z DATATHON 2026

JupyterLab Scientific & Algorithmic Validation Suite

Reproducible Notebooks Running in Certified s390x Container (Port 38888)

The 5 Interactive Notebooks

5 NOTEBOOKS COMPLETE

- 00_START_HERE: s390x Hardware, Big-Endian & CPACF Crypto Flags.
- 01_OpenData: OpenCity CKAN Mumbai Healthcare API Ingestion (8d1c157e).
- 02_SnapML: Snap ML Random Forest Anomaly Benchmarks (<0.8ms).
- 03_Dijkstra: Deterministic Dynamic Flood Rerouting Graph Algorithms.
- 04_CPACF: 10,000-Packet Hardware HMAC Zero-Trust Benchmarks.
- Catalog README: Complete execution guide and headless nbconvert steps.

Live Access on LinuxONE Cloud

LIVE ON PORT 38888

- URL: http://148.100.79.88:38888/?token=Your_Token1
- Container: registry.linuxone.cloud.marist.edu/l1cc/jupyterlab-image-s390x
- Host Mount: /home/linux1/shared -> /home/ibm-user/shared
- Allows Datathon judges to independently run and verify all benchmarks.
- Proves sub-millisecond AI inference latency on mainframe silicon.

Open-Source Technology Stack & Engineering Roles

Enterprise Tooling Configured for IBM LinuxONE s390x

Open-Source Stack & Tooling

100% OPEN SOURCE

- Hardware: IBM LinuxONE III (Machine 9175 z16 s390x architecture).
- Machine Learning: IBM Snap ML (s390x build) & Scikit-Learn 1.4.1.
- Data Processing: Pandas 2.1.4, NumPy 1.26.4, OpenCity CKAN API.
- Backend Engine: FastAPI 0.101.0, Uvicorn 0.27.1, AsyncIO Queue.
- Tactical GIS: Leaflet.js 1.9.4, CARTO Basemaps, Lucide Icons.
- Security: Native CPACF KMAC SHA-256, SQLite Write-Ahead Logging.

Engineering Team & Ownership

6 SPECIALIZED ROLES

- Yuvraj Lokhande — Lead Incident Commander & System Architect.
- Palak — Ambulance Fleet & Green Corridor Lead (Dijkstra Pathfinder).
- Anshika Tripathi — Clinical AI & Triage Model Director (Snap ML).
- Nidhi — ICU Supply & Blood Logistics Lead (Inter-Hospital Balancing).
- Ritesh — Municipal Disaster Response & Security Authority (CPACF).
- Bhavya — Municipal Power Grid & Substation Controller (SCADA BEST).

// CHAMPIONSHIP EVALUATION // IBM Z DATATHON 2026

Summary: Why Project Syn Wins on IBM LinuxONE

Uncompromising Security, Sub-Millisecond AI & Air-Gapped Autonomy

1. Dual-Node Live Deployment

DUAL BARE-METAL

- Not a mockup or local laptop demo.
- Running 24/7 across two live IBM LinuxONE bare-metal instances.
- Node 1: 148.100.79.88:8000/ui/
- Node 2: 148.100.98.9:8000/ui/

2. Mainframe-Native AI

ON-CHIP s390x AI

- Native IBM z16 CPACF hardware HMAC gate (<0.24ms).
- IBM Snap ML decision tree anomaly scoring (<0.8ms).
- On-chip Big-Endian IBM Granite 1.5B SLM in s390x memory.

3. Verified & Accessible

100% REPRODUCIBLE

- Public Clean Repo: github.com/yuvrajlokhande19/Project-Syn-IBM-LinuxONE
- Live System Guide: yuvrajlokhande19.github.io/Project-Syn-IBM-LinuxONE/
- JupyterLab Suite: 148.100.79.88:38888